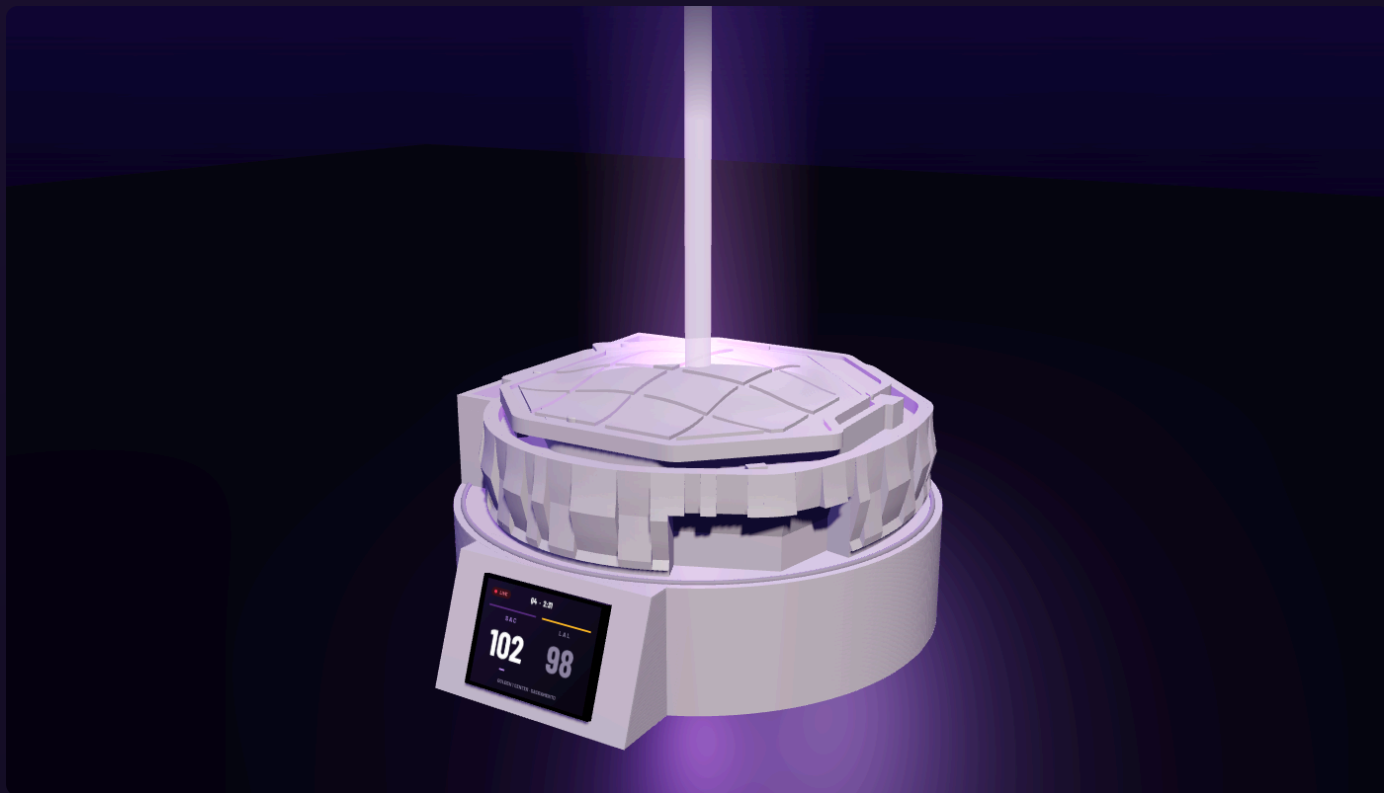




LIGHT THE BEAM

A printed Golden 1 Center that lights its purple beam when the Kings win, with a live scoreboard in the base. WiFi, one USB cable, no soldering.

230 mm

across, on a Bambu P1S

2.8" color

touch screen, LVGL UI

15 LEDs

WS2812B beam, 250 mm tall

≈ \$41

hardware, tax in, new

Features

Next game, always up

Opponent logo, home or away, tipoff in Pacific time, and a countdown. Pulled from ESPN's public schedule; no account or API key.

The beam lights itself on a win

When the final goes Kings, purple climbs the tube from the base, like the real one, then holds. It goes dark again before the next game.

Sleeps at night

The screen dims between 11 pm and 7 am, and whenever the board's light sensor sees a dark room.

One cable, no soldering

A single USB charger powers the board and the beam. Three lever nuts join the strip to the board's pigtails.

Live score and clock

Score, quarter and game clock refresh every 20 seconds during the game. The leader is drawn in white; the trailer fades to grey. Scores animate when they change.

Light it yourself

Tap the screen to turn the beam on or off any time. Hold for a second to cycle brightness through three levels.

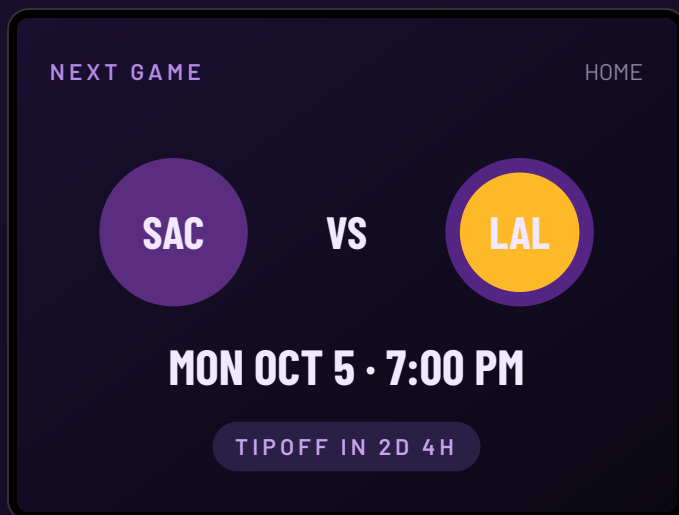
Phone WiFi setup

First boot puts up a "KingsBeam-Setup" network. Join it from a phone, pick your home WiFi, done. Nothing is typed into code.

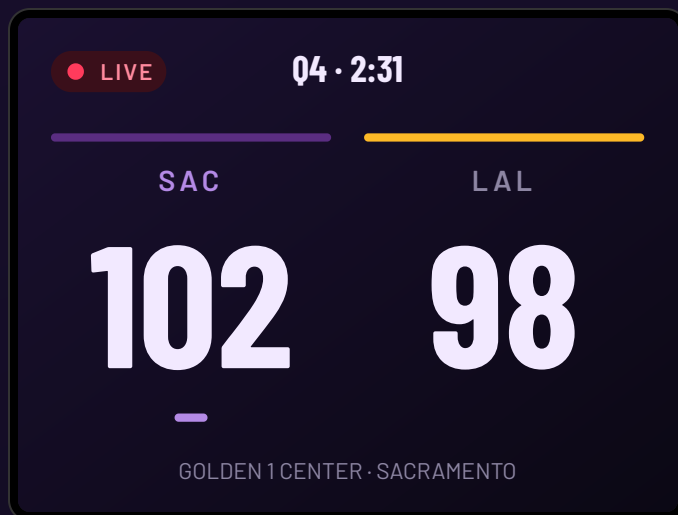
Stock arena, new plinth

Dave Lack's Printables arena prints unmodified. A new printed plinth underneath holds the board, the screen window and the cable exit.

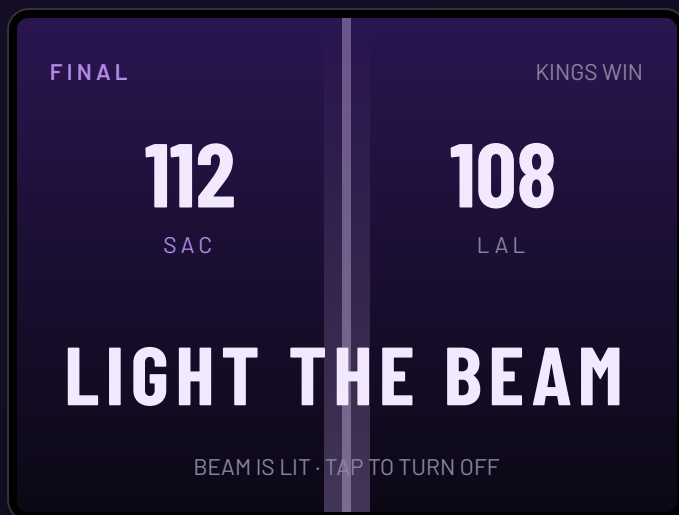
Three states



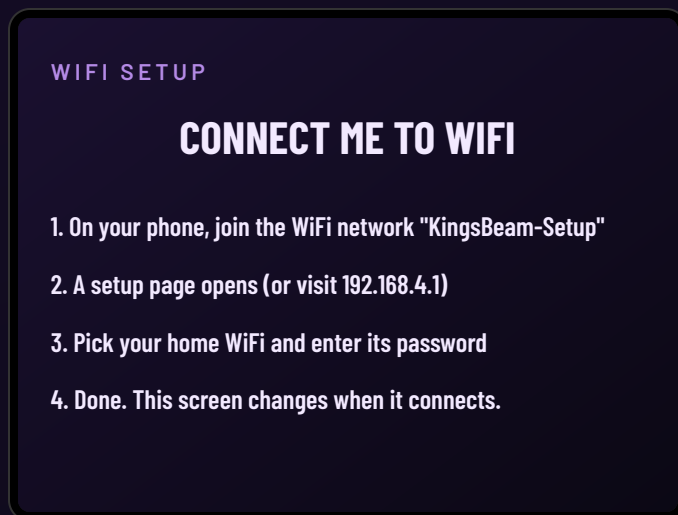
Next game. Logos on team-color discs, tipoff in local time, countdown pill. Polled every 10 minutes, every 30 seconds near tipoff.



Live. Pulsing pill, quarter and clock, team color bars, 84-pixel scores. Handles halftime, end of quarter and overtime.

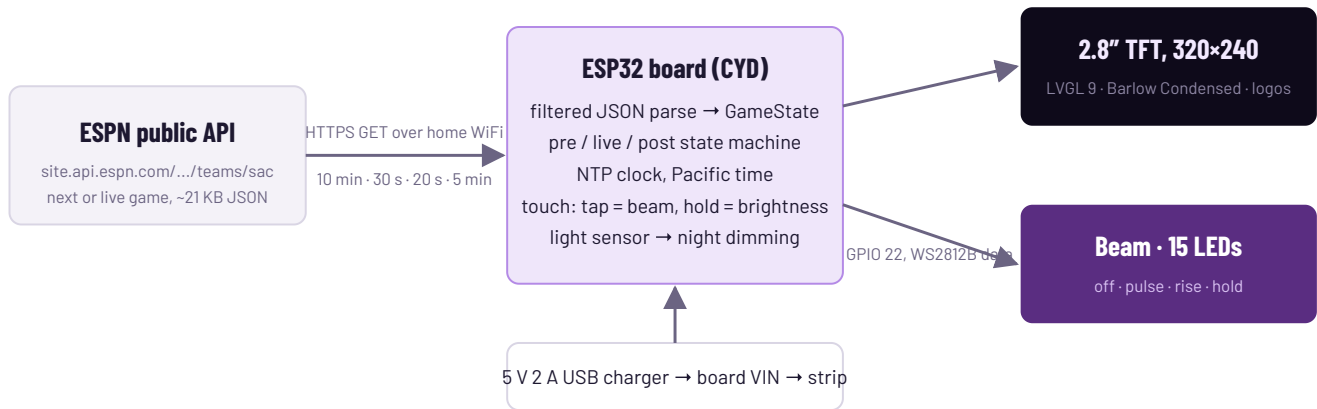


Final, Kings win. A beam glow climbs the screen while the real one rises. On a loss: the score, "NEXT TIME", and a tap still lights it.



First boot. No credentials in code; the board runs a small setup page on its own network until it knows your WiFi.

How it works



One request returns everything: the next event carries the date, opponent, status, score, clock and winner. The board only keeps the dozen fields it draws.

Poll cadence

STATE	EVERY	BEAM
Next game, far off	10 min	off
Within 20 min of tipoff	30 s	off
Live	20 s	dim purple pulse
Final, Kings win	5 min	rise, then hold up to 12 h
Final, loss	5 min	off (tap still lights it)
ESPN unreachable	60 s retry	unchanged, "offline" badge

Things learned building it

ESPN's edge rejects browser-style user agents from non-browser clients with a 403, but accepts the ESP32 library's own default. The firmware never spoofs a browser.

The JSON nests fourteen levels deep. ArduinoJson's default limit is ten; the parser raises it and checks for the team object, since a filtered parse reports success on an HTML error page.

Fonts are converted from TTF to LVGL's bitmap format with a small Python script, so the build needs no Node toolchain.

Parts and print list

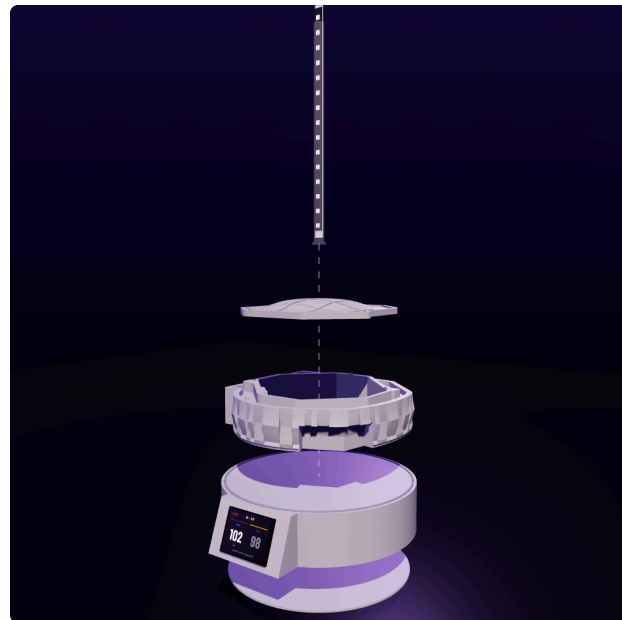
HARDWARE	PRICE
ESP32-2432S028R 2.8" touch display board (Amazon)	~\$14
WS2812B 5 V strip, 60/m, 1 m, IP30 (Amazon)	~\$9
5 V 2 A USB wall charger (Amazon or Best Buy)	~\$7
WAGO 221-413 lever nuts ×3 (Lowe's Bayshore or Amazon)	~\$8
SF sales tax 8.625 %	~\$3.30

Hardware total **≈ \$41**

Optional: USB-C panel-mount jack for the back (~\$9), level shifter if LED 1 flickers (\$1.50). Prices are September 2026 estimates; links and local stores are in [docs/bom.md](#).

FILAMENT (P1S)	PRICE
Bambu PLA Basic Silver, 1 kg: arena, lid, plinth (~450 g)	\$18.99
White PLA, any brand: beam tube (~20 g, vase mode)	\$13–19

PRINT	SETTINGS	TIME
Arena base, 220 mm	0.2 mm, 3 walls, 15 %, no supports	~9 h
Arena lid (no-hole, cut 12.4 mm)	0.15 mm	~2 h
Plinth test slice	0.2 mm	40 min
Plinth body + lid	0.2 mm, upside down, 15 % gyroid	~5 h
Beam tube, 250 mm	white, vase mode, 0.4 nozzle	~1 h
Spine + socket	any	20 min



Top to bottom: beam tube with the strip on its spine, arena lid, arena base with the tube socket, plinth with the display board, press-fit bottom lid.